JC4002 Knowledge Representation

Day 8: Modern Description Logic

Dr Ferdian Jovan	ferdian.jovan@abdn.ac.uk
Dr Aiden Durrant	aiden.durrant@abdn.ac.uk
Dr Jiangtian Nie	jiangtian.nie@abdn.ac.uk

Day 8
Modern Description Logic

1. **Modern DL**
2. Attributive Concept Language with Complements ($\mathcal{ALC}$)
3. $\mathcal{ALC}$ Reasoning Services

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Modern Notation

"A man that is married to a doctor, and all of whose children are either doctors or professors."

- With the notation we learnt in the previous lecture (called Baader&Lutz notation):

  [**AND** Man
  [**EXISTS** :married Doctor]
  [**ALL** :hasChild [**OR** Doctor Professor]]

Modern Notation

"A man that is married to a doctor, and all of whose children are either doctors or professors."

- With the notation we learnt in the previous lecture (called Baader&Lutz notation):

  [**AND** Man
  [**EXISTS** :married Doctor]
  [**ALL** :hasChild [**OR** Doctor Professor]]

- This notation is less intuitive, not very clear, and not standardised.
- Current notation uses symbols like FOL for clarity and standardisation.

  $Human \sqcap \neg Female \sqcap (\exists married.Doctor) \sqcap (\forall hasChild.(Doctor \sqcup Professor))$.

- With this standardisation, DL has become a fundamental part of the Semantic Web, through the development **Web Ontology Language** (OWL).

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Basics of Modern DL

Let's redefine (proto-)Description-Logic DL as a tuple (Sym_{DL}, For_{DL}, Int_{DL}, $\models_{DL}$).

- *Symbols* of DL (Sym_{DL}) = B ⋃ C ⋃ R ⋃ U
    - B = {'(', ')', ',', ⊑},
    - C are all *atomic concepts* (e.g. Father, Mother),
    - R are all *roles / relations* (e.g. hasChild, owns),
    - U are all *constants / individuals* (e.g. john, mary),
    - where $\mathrm{B} \cap C \cap \mathrm{R} \cap \mathrm{U}$ = {} (an empty set).

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Basics of Modern DL

Let's redefine (proto-)Description-Logic DL as a tuple (Sym_{DL}, For_{DL}, Int_{DL}, $\vDash_{DL}$).

- *Symbols* of DL (Sym_{DL}) = B ∪ C ∪ R ∪ U
    - B = {'(', ')', ',', ⊑},
    - C are all *atomic concepts* (e.g. Father, Mother),
    - R are all *roles / relations* (e.g. hasChild, owns),
    - U are all *constants / individuals* (e.g. john, mary),
    - where B ∩ *C* ∩ R ∩ U = ∅ (an empty set).
- *Formulas / Sentences* of DL (For_{DL})
    - If d ∈ C and a ∈ U, then d(a) ∈ For_{DL}(e.g. Father(john), Mother(mary));
    - If r ∈ R, and c_1, c_2 ∈ U, then r(c_1, c_2) ∈ For_{DL}(e.g. hasChild(john, mary));
    - if d_1 and d_2 ∈ C, then ($d_1 \sqsubseteq d_2$) ∈ For_{DL} (e.g. Father ⊑ Person);

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Basics of Modern DL

Let's redefine (proto-)Description-Logic DL as a tuple (Sym_{DL}, For_{DL}, Int_{DL}, $\vDash_{DL}$).

- An *interpretation* in DL (Int_{DL}) is a pair $\zeta = (\delta, \varphi)$ such that:
    - δ is a non-empty set called the *domain* of interpretation;
    - φ is a function defined on the sets of concepts, roles, and individuals such that:
        - for each $c \in C$, $\varphi(c) \subseteq \delta$;
        - for each $a \in U$, $\varphi(a) \in \delta$;
        - for each $r \in R$, $\varphi(r) \subseteq \delta \times \delta$.

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Basics of Modern DL

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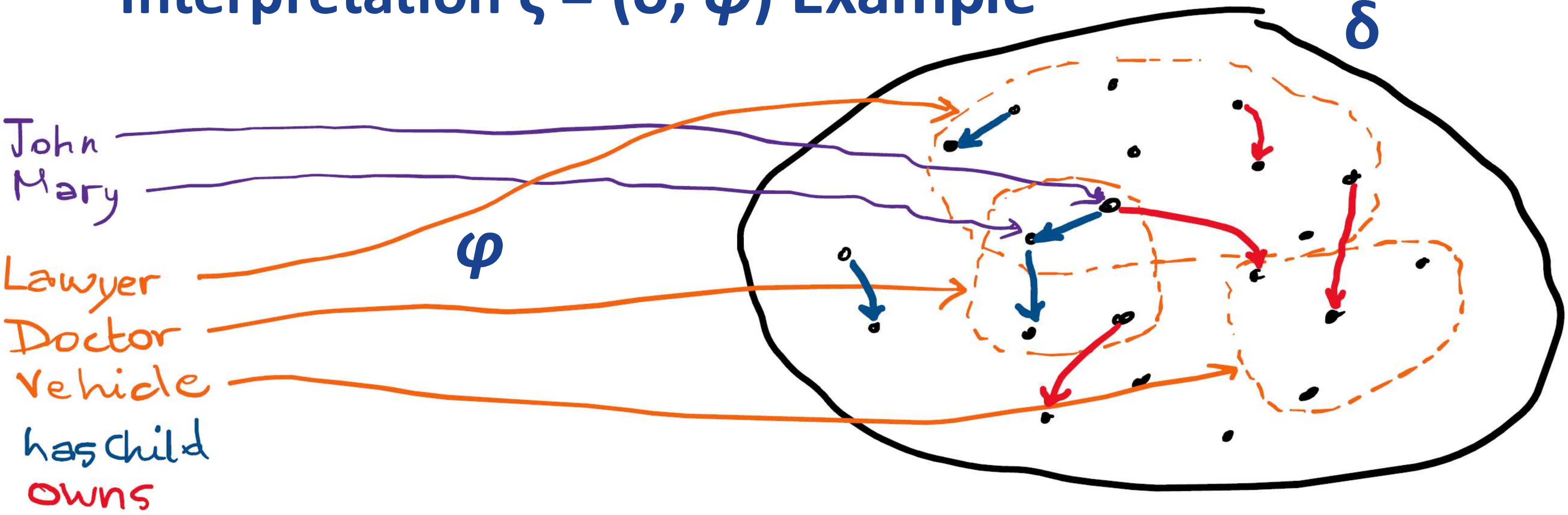
- An *interpretation* in DL (Int_{DL}) is a pair $\zeta = (\delta, \varphi)$ such that:
  - δ is a non-empty set called the *domain* of interpretation;
  - φ is a function defined on the sets of concepts, roles, and individuals such that:
    - for each $c \in C$, $\varphi(c) \subseteq \delta$;
    - for each $a \in U$, $\varphi(a) \in \delta$;
    - for each $r \in R$, $\varphi(r) \subseteq \delta \times \delta$.
- An interpretation $\zeta = (\delta, \varphi)$ in DL *satisfies* ($\vDash_{DL}$):

1. $a \sqsubseteq b$ iff $\varphi(a) \subseteq \varphi(b)$;
2. $c(a)$ iff $\varphi(a) \in \varphi(c)$;
3. $r(a, b)$ iff $(\varphi(a), \varphi(b)) \in \varphi(r)$.

Interpretation ζ = (δ, φ) Example



This interpretation satisfies: Lawyer(John), Lawyer(Mary), Doctor(John), Doctor(Mary), and hasChild(John,Mary), Doctor ⊑ Doctor, Lawyer ⊑ Lawyer, Vehicle ⊑ Vehicle

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ABox and TBox

- In a DL theory, formulas can be partitioned into two distinguished sets: one that contains all *concept inclusions (TBox)* and one that contains all *concept assertions* and *roles assertions (ABox)*.
- A **TBox** contains the terminological knowledge:
  - Concept definitions (introduce names for concepts):

    $$\text{Father} \doteq \text{Man} \sqcap \exists \text{haschild}.\top$$

  - Axioms (restrict models):

    $$\text{Mother} \sqsubseteq \text{Woman}$$

    $$\text{BlackCat} \sqsubseteq \text{Cat} \sqcap \forall \text{hascolour}.\text{Black}$$

- Note that $\doteq$, $\sqcap$, $\forall$, $\exists$ **are not part of the proto-DL** we just defined earlier (because they are not part of Sym_{DL}). They are here for clarity purposes.

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ABox and TBox

- In a DL theory, formulas can be partitioned into two distinguished sets: one that contains all *concept inclusions (TBox)* and one that contains all *concept assertions* and *roles assertions (ABox)*.
- A **ABox** contains assertions about individuals:
    - Concept assertions:

      BlackCat(mimi)

      (Mother ⊓ ∃haschild.Woman)(mimi)
    - Role assertions:

      haschild(mary, betty)
- Note that ≐, ⊓, ∀, ∃ **are not part of the proto-DL** we just defined earlier (because they are not part of Sym_{DL}). They are here for clarity purposes.

The Description Logic $\mathcal{FL}_0$

We define the logic $\mathcal{FL}_0$ as a tuple ($Sym_{\mathcal{FL}0}$, $For_{\mathcal{FL}0}$, $Int_{\mathcal{FL}0}$, $\vDash_{\mathcal{FL}0}$) where:

- $Sym_{\mathcal{FL}0} = Sym_{DL} \cup Cons_{\mathcal{FL}0}$
  - $Cons_{\mathcal{FL}0} = \{\sqcap, \forall, \text{'.'}, \top\}$;
  - Sym_{DL} is as defined in proto-DL;
  - and $Sym_{DL} \cap Cons_{\mathcal{FL}0}$ are disjoint.
- The set of complex concepts is the minimal set containing the following:
  - Every atomic concept A ∈ C is a complex concept;
  - $\top$ is a complex concept;
  - For all complex concepts c and d, ($c \sqcap d$) is a complex concept;
  - For all $r \in R$, and all complex concepts c, $\forall r.c$ is a complex concept.

The Description Logic $\mathcal{FL}_0$

We define the logic $\mathcal{FL}_0$ as a tuple ($Sym_{\mathcal{FL}0}$, $For_{\mathcal{FL}0}$, $Int_{\mathcal{FL}0}$, $\vDash_{\mathcal{FL}0}$) where:

- *Formulas* of $\mathcal{FL}_0$ ($For_{\mathcal{FL}0}$):
  - For all complex concept d and a ∈ U, then d(a) ∈ $For_{\mathcal{FL}0}$ (these formulas are called **concept assertions** as part of ABox);
  - If r ∈ R, and c_1, c_2 ∈ U, then r(c_1, c_2) ∈ $For_{\mathcal{FL}0}$ (these formulas are called **role assertions** as part of ABox);
  - For all complex concepts d_1 and d_2, ($d_1 \sqsubseteq d_2$) ∈ $For_{\mathcal{FL}0}$ (these formulas are called **general concept inclusions** or **GCIs**);
- An *interpretation* in $\mathcal{FL}_0$ ($Int_{\mathcal{FL}0}$) is similar to Int_{DL} with extensions for:
  - $\varphi(T) = \delta$;
  - For all complex concepts c and d, $\varphi(c \sqcap d) = \varphi(c) \cap \varphi(d)$;
  - For all r ∈ R, and all complex concepts c, $\varphi(\forall r.c) = \{x \in \delta \mid$ for any $y \in \delta$ s.t. $(x, y) \in \varphi(r)$ and $y \in \varphi(c)\}$

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The Description Logic $\mathcal{FL}_0$

We define the logic $\mathcal{FL}_0$ as a tuple ($Sym_{\mathcal{FL}0}$, $For_{\mathcal{FL}0}$, $Int_{\mathcal{FL}0}$, $\vDash_{\mathcal{FL}0}$) where:

- An *interpretation* in $\mathcal{FL}_0$ ($Int_{\mathcal{FL}0}$) is similar to Int_{DL} with extensions for:
  - $\varphi(T) = \delta$;
  - For all complex concepts c and d, $\varphi(c \sqcap d) = \varphi(c) \cap \varphi(d)$;
  - For all $r \in R$, and all complex concepts c, $\varphi(\forall r.c) = \{x \in \delta \mid$ for any $y \in \delta$ s.t. $(x, y) \in \varphi(r)$ and $y \in \varphi(c)\}$
- An interpretation $\zeta = (\delta, \varphi)$ in $\mathcal{FL}_0$ *satisfies* ($\vDash_{\mathcal{FL}0}$):
  1. a *general concept inclusion* $a \sqsubseteq b$ iff $\varphi(a) \subseteq \varphi(b)$;
  2. a *concept assertion* $c(a)$ iff $\varphi(a) \in \varphi(c)$;
  3. a *role assertion* $r(a, b)$ iff $(\varphi(a), \varphi(b)) \in \varphi(r)$.

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The Description Logic $\mathcal{FL}_0$

With Logic $\mathcal{FL}_0$, we can represent some complex concepts we encountered earlier:

- $\text{BlackCat} \sqsubseteq \text{Cat} \sqcap \forall \text{hascolour}.\text{Black}$
- $\text{Person} \sqsubseteq \forall \text{hasParent}.\text{Person}$
- $T \sqsubseteq (C \sqcap \forall P.(D \sqcap D))$

The Description Logic $\mathcal{FL}_0$

With Logic $\mathcal{FL}_0$, we can represent some complex concepts we encountered earlier:

- BlackCat ⊑ Cat ⊓ ∀hascolour.Black
- Person ⊑ ∀hasParent.Person
- T ⊑ (C ⊓ ∀P.(D ⊓ D))

But you still can't represent *complement / negation* and *existensial*.

- Human ⊓ ¬Female ⊓ (∃married.Doctor) ⊓ (∀hasChild.(Doctor ⊔ Professor)))

We need DL that is more expressive than $\mathcal{FL}_0$.

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The Description Logic $\mathcal{FL}_\perp$ & $\mathcal{FL}^-$

We define the logic $\mathcal{FL}_\perp$ as a tuple ($Sym_{\mathcal{FL}\perp}$, $For_{\mathcal{FL}\perp}$, $Int_{\mathcal{FL}\perp}$, $\vDash_{\mathcal{FL}\perp}$) where:

- $Sym_{\mathcal{FL}\perp} = Sym_{\mathcal{FL}0} \cup \{\perp\}$, assuming $\perp$ is disjoint from $Sym_{\mathcal{FL}0}$;
- $\perp$ is another complex concept and can be used in any syntactic constructions of $\mathcal{FL}_0$ where a complex concept can be used;
- *Formulas* of $\mathcal{FL}_\perp$ ($For_{\mathcal{FL}\perp}$) is defined identically as $For_{\mathcal{FL}0}$, except that complex concepts can be any of $\mathcal{FL}_\perp$ rather than just those of $\mathcal{FL}_0$.
- $Int_{\mathcal{FL}\perp} = Int_{\mathcal{FL}0}$ with the addition $\varphi(\perp) = \emptyset$ ($\perp$ is always interpreted as the empty set);
- The satisfaction relation $\vDash_{\mathcal{FL}\perp}$ is defined as in $\mathcal{FL}_0$.

With logic $\mathcal{FL}_\perp$, we can now define Father $\sqcap$ Mother $\sqsubseteq \perp$.

The Description Logic $\mathcal{FL}_\perp$ & $\mathcal{FL}^-$

We define the logic $\mathcal{FL}^-$ as a tuple ($Sym_{\mathcal{FL}^-}$, $For_{\mathcal{FL}^-}$, $Int_{\mathcal{FL}^-}$, $\vDash_{\mathcal{FL}^-}$) where:

- $Sym_{\mathcal{FL}^-} = Sym_{\mathcal{FL}\perp} \cup \{\neg\}$, assuming $\neg$ is disjoint from $Sym_{\mathcal{FL}\perp}$;
- For any atomic concept A $\in$ C, $\neg$A is another complex concept and can be used in any syntactic constructions of $\mathcal{FL}_\perp$ where a complex concept can be used;
- *Formulas* of $\mathcal{FL}^-$ ($For_{\mathcal{FL}^-}$) is defined identically as $For_{\mathcal{FL}\perp}$, except that complex concepts can be any of $\mathcal{FL}^-$ rather than just those of $\mathcal{FL}_\perp$.
- $Int_{\mathcal{FL}\perp} = Int_{\mathcal{FL}0}$ with the addition $\varphi(\neg A) = \delta \setminus \varphi(A)$ ($\neg$A is interpreted as the complement of $\varphi(A)$ in δ);
- The satisfaction relation $\vDash_{\mathcal{FL}^-}$ is defined as in $\mathcal{FL}_\perp$.

Note that the symbol $\neg$ can only be used in front of a concept name (atomic concept), **not before** an arbitrary **complex concept**.

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The Description Logic $\mathcal{FL}^-$

With Logic $\mathcal{FL}^-$, we can represent some complex concepts we encountered earlier:

- ❑ BlackCat ⊑ Cat ⊓ ∀hascolour.Black
- ❑ Person ⊑ ∀hasParent.Person
- ❑ T ⊑ (¬C ⊓ ∀P.(D ⊓ D))

But you still can't represent:

- ❑ Human ⊓ ¬Female ⊓ (∃married.Doctor) ⊓ (∀hasChild.(Doctor ⊔ Professor)))

So what are still missing to represent the above formula?

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The Description Logic $\mathcal{AL}$

We define the logic $\mathcal{AL}$ as a tuple ($Sym_{\mathcal{AL}}$, $For_{\mathcal{AL}}$, ,$Int_{\mathcal{AL}}$, $\vDash_{\mathcal{AL}}$) where:

- $Sym_{\mathcal{AL}} = Sym_{\mathcal{FL}^-} \cup \{\exists\}$, assuming $\exists$ is disjoint from $Sym_{\mathcal{FL}^-}$;
- For any role r ∈ R and any concept c, ∃r.c is another complex concept and can be used in any syntactic constructions of $\mathcal{FL}^-$ where a complex concept can be used;
- *Formulas* of $\mathcal{AL}$ ($For_{\mathcal{AL}}$) is defined identically as $For_{\mathcal{FL}^-}$, except that complex concepts can be any of $\mathcal{AL}$ rather than just those of $\mathcal{FL}^-$.
- $Int_{\mathcal{AL}} = Int_{\mathcal{FL}^-}$ with the addition $\varphi(\exists r.c) = \{x \in \delta \mid \text{there is } y \in \varphi(c) \text{ s.t. } (x, y) \in \varphi(r)\}$;
- The satisfaction relation $\vDash_{\mathcal{AL}}$ is defined as in $\mathcal{FL}^-$.

The Description Logic $\mathcal{AL}$

With Logic $\mathcal{AL}$, we can represent some complex concepts we encountered earlier:

- BlackCat ⊑ Cat ⊓ ∀hascolour.Black
- Person ⊑ ∀hasParent.Person
- ⊤ ⊑ (¬C ⊓ ∀P.(D ⊓ D))
- Human ⊓ ¬Female ⊓ (∃married.Doctor) ⊓ (∀hasChild.(Doctor)))

$\mathcal{AL}$ is often considered as the archetypical description logic, from which others are defined.

Each construct is assigned a letter that can be concatenated to $\mathcal{AL}$ to form the name of a particular DL (for instance $\mathcal{ALCIO}$ combines the constructs of $\mathcal{AL}$ with general concept complement -- $\mathcal{C}$, inverse roles -- $\mathcal{I}$, and nominals -- $\mathcal{O}$).

What are those constructs?

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$\mathcal{AL}$ Extension

We can extend the $\mathcal{AL}$ family with some classical construct to make the logic more expressive.

- **concept constructs** are new syntactic features that can make new kinds of complex concepts.

Symbol	Name	Syntax	Semantics
Concept constructs			
$\mathcal{A}$	Value restriction (already in $\mathcal{AL}$)	$\forall r.c$	$\{x \in \Delta \mid \text{for all } y \in \Delta \text{ such that } (x, y) \in \mathcal{I}(r), y \in \mathcal{I}(c)\}$
$\mathcal{E}$	General existential quantification	$\exists r.c$	$\{x \in \Delta \mid \text{there exists } y \in c \text{ such that } (x, y) \in \mathcal{I}(r)\}$
$\mathcal{U}$	Concept disjunction	$c \sqcup d$	$\mathcal{I}(c) \cup \mathcal{I}(c)$
$\mathcal{C}$	General concept complement	$\neg c$	$\Delta \setminus \mathcal{I}(c)$
$\mathcal{O}$	Nominal	$\{a\}$	$\{\mathcal{I}(a)\}$
$\mathcal{N}$	Cardinality restriction	$\leq n.r$	$\{x \in \Delta \mid \text{card}(\{y \in \Delta \mid (x, y) \in \mathcal{I}(r)\}) \leq n$
		$\geq n.r$	$\{x \in \Delta \mid \text{card}(\{y \in \Delta \mid (x, y) \in \mathcal{I}(r)\}) \geq n$
$\mathcal{Q}$	Qualified cardinality restriction	$\leq n.r.c$	$\{x \in \Delta \mid \text{card}(\{y \in \mathcal{I}(c) \mid (x, y) \in \mathcal{I}(r)\}) \leq n\}$
		$\geq n.r.c$	$\{x \in \Delta \mid \text{card}(\{y \in \mathcal{I}(c) \mid (x, y) \in \mathcal{I}(r)\}) \geq n\}$

$\mathcal{AL}$ Extension

We can extend the $\mathcal{AL}$ family with some classical construct to make the logic more expressive.

❑ **role constructs** are syntactic features than can make complex roles.

Symbol	Name	Syntax	Semantics
Role constructs			
$\mathcal{I}$	Inverse role	r^-	$\{(x, y) \in \Delta \times \Delta \mid (y, x) \in \mathcal{I}(r)\}$
$(\sqcap)$	Role conjunction	$r \sqcap_r s$	$\mathcal{I}(r) \cap \mathcal{I}(s)$
$(\sqcup)$	Role disjunction	$r \sqcup_r s$	$\mathcal{I}(r) \cup \mathcal{I}(s)$
$(\neg)$	Role complement	$\neg_r r$	$\Delta \times \Delta \setminus \mathcal{I}(r)$

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$\mathcal{AL}$ Extension

We can extend the $\mathcal{AL}$ family with some classical construct to make the logic more expressive.

- **axiom constructs** are new types of formulas / sentences.

Symbol	Name	Syntax	Semantics
Additional axioms			
$\mathcal{F}$	Role functionality	$\top \sqsubseteq \leq 1.r$	if $(x, y) \in \mathcal{I}(r)$ and $(x, z) \in \mathcal{I}(r)$ then $x = z$
$\mathcal{H}$	Role hierarchy	$r \sqsubseteq_{\mathrm{r}} s$	$\mathcal{I}(r) \subseteq \mathcal{I}(s)$
$\mathcal{S}$	$\mathcal{ALC}$ + Role transitivity	Trans(r)	if $(x, y) \in \mathcal{I}(r)$ and $(y, z) \in \mathcal{I}(r)$ then $(x, z) \in \mathcal{I}(r)$
$\mathcal{R}$	Role chain	$r_1 \circ \ldots \circ r_n \sqsubseteq_{\mathrm{r}} s$	if $(x, y_1) \in \mathcal{I}(r_1)$, …, $(y_i, y_{i+1}) \in \mathcal{I}(r_{i+1})$, … then $(x, y_n) \in \mathcal{I}(s)$

Note that $\mathcal{S}$ is used as a shortcut for $\mathcal{ALC}$ + transitivity, so the logic $\mathcal{SHI}$ denotes the DL that has general complement, role transitivity, role hierarchy, and role inverse.

- $\mathcal{SHIOQ}$ is the basis of OWL, $\mathcal{SRIOQ}$ is the basis of OWL 2.0

Activity 1

In pairs, discuss the following question (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):

"Which extension does the $\mathcal{AL}$ need to express

$$\text{Human} \sqcap \neg\text{Female} \sqcap (\exists\text{married}.\text{Doctor}) \sqcap \neg(\forall\text{hasChild}.(\text{Doctor} \sqcup \text{Professor}))$$**?"**

Attributive concept Language with Complements ($\mathcal{ALC}$) logic

- Can $\mathcal{AL}$ Logic represent all complex concepts?

$$\neg(\text{Human} \sqcap \text{Female}) \sqcap \neg(\exists \text{married}.\text{Doctor})$$

- Although $\neg$ has been introduced in $\mathcal{FL}^-$ but $\neg$ is limited to **atomic concept**
- An extension to $\mathcal{AL}$ is needed to have more expressive $\neg$.
- Concept construct $\mathcal{C}$ is a perfect fit for this. It generalises $\neg$ to cover **complex concepts** as well.
- Some constructs subsume others. In particular, $\mathcal{Q}$ subsumes $\mathcal{N}$, $\mathcal{N}$ subsumes $\mathcal{F}$, $\mathcal{EC}$ subsumes $\mathcal{A}$, $\mathcal{AC}$ subsumes $\mathcal{E}$, $\mathcal{ALC}$ subsumes $\mathcal{UE}$, $\mathcal{R}$ subsumes $\mathcal{SH}$, $(\neg,\cap)$ subsumes $(\cup)$, and $(\neg,\cup)$ subsumes $(\cap)$.
- As $\mathcal{ALC}$ subsumes $\mathcal{UE}$, a concept of disjunction $(\sqcup)$ is covered by $\mathcal{ALC}$.

$\mathcal{ALC}$ Concepts (Summary)

- ❑ $\mathcal{ALC}$ is the basic description logic
- ❑ $\mathcal{ALC}$ complex concepts C are inductively defined from atomic concept A and roles r:

$$C ::= \top \mid \bot \mid A \mid \neg C \mid C \sqcap D \mid C \sqcup D \mid \forall r.C \mid \exists r.C$$

- ❑ Semantics given through interpretations $\zeta = (\delta, \varphi)$ with
  - $\varphi(\top) = \delta$;
  - $\varphi(\bot) = \emptyset$;
  - $\varphi(\neg C) = \delta \setminus \varphi(C)$;
  - $\varphi(C \sqcap D) = \varphi(C) \cap \varphi(D)$;
  - $\varphi(C \sqcup D) = \varphi(C) \cup \varphi(D)$;
  - $\varphi(\exists r.C) = \{x \in \delta \mid \text{there is } y \in \varphi(C) \text{ s.t. } (x, y) \in \varphi(r)\}$;
  - $\varphi(\forall r.C) = \{x \in \delta \mid \text{for any } y \in \delta \text{ s.t. } (x, y) \in \varphi(r) \text{ and } y \in \varphi(C)\}$

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$\mathcal{ALC}$ Knowledge Base (ABox, TBox)

- An $\mathcal{ALC}$ Knowledge Base is a pair $\Sigma = (\boldsymbol{T}, \boldsymbol{A})$, where
  - $\boldsymbol{T}$ is a TBox
  - $\boldsymbol{A}$ is a ABox
- An interpretation ζ is a *model* of Σ if it *satisfies* $\boldsymbol{T}$ and $\boldsymbol{A}$.
- An interpretation ζ satisfies TBox $\boldsymbol{T}$ iff it satisfies all elements of $\boldsymbol{T}$ including
  - $C \sqsubseteq D$ iff $\varphi(C) \subseteq \varphi(D)$;
  - $C \doteq D$ iff $\varphi(C) \subseteq \varphi(D)$ and $\varphi(D) \subseteq \varphi(C)$.
- An interpretation ζ satisfies ABox $\boldsymbol{A}$ iff it satisfies all elements of $\boldsymbol{A}$ including
  - $C(a)$ iff $\varphi(a) \in \varphi(C)$;
  - $r(a, b)$ iff $(\varphi(a), \varphi(b)) \in \varphi(r)$.
- Note that the semantics of ABox and TBox is equal to $\vDash_{\mathcal{FL}0}$ definition.

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$\mathcal{ALC}$ Knowledge Base Reasoning

- Given an $\mathcal{ALC}$ KB $\Sigma = (\boldsymbol{T}, \boldsymbol{A})$, we may want to answer questions about individuals and/or KB as a whole:
  - Is KB consistent, i.e., does there exist a model (interpretation)?
  - Will KB still be consistent if we add new concepts?
  - Can we infer additional information about individuals?
- All these reasonings boil down to entailments and logical consequences
  - The same concept introduced in FOL.
  - Given a formula / sentence S, $\Sigma \models S$ iff every model of Σ is a model of S.
- $\boldsymbol{T} = \{\exists teaches.Course \sqsubseteq GraduateStudent \sqcup Professor\}$
- $\boldsymbol{A} = \{teaches(john, cs101), Course(cs101), \neg Professor(john)\}$

$\mathcal{ALC}$ Knowledge Base Reasoning

- Given an $\mathcal{ALC}$ KB $\Sigma = (\boldsymbol{T}, \boldsymbol{A})$, we may want to answer questions about individuals and/or KB as a whole:
  - Is KB consistent, i.e., does there exist a model (interpretation)?
  - Will KB still be consistent if we add new concepts?
  - Can we infer additional information about individuals?
- All these reasonings boil down to entailments and logical consequences
  - The same concept introduced in FOL.
  - Given a formula / sentence S, Σ |= S iff every model of Σ is a model of S.
- $\boldsymbol{T} = \{\exists \text{teaches}.\text{Course} \sqsubseteq \text{GraduateStudent} \sqcup \text{Professor}\}$
- $\boldsymbol{A} = \{\text{teaches}(\text{john}, \text{cs101}), \text{Course}(\text{cs101}), \neg\text{Professor}(\text{john})\}$
- $\Sigma \models \text{GraduateStudent}(\text{john})$

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3. **$\mathcal{ALC}$ Reasoning Services**

Basic Reasoning Problems and Services

- What kinds of reasoning problems and services might be interesting?
- Scenario: Ontology design
    - We are building a conceptual model (a TBox) for our domain
    - At this design stage we haven't yet included the data (no ABox)
- Our TBox should be
    - Error-free: No unintended logical consequences
    - Sufficiently detailed: Contain all relevant knowledge for our application

Ontology Design

TBox $\boldsymbol{T}$ = {JuvArthritis ⊑ Arthritis ⊓ JuvDisease,

Arthritis ⊑ ∃damages.Joint ⊓ ∀damages.Joint ⊓ ∃affects.Adult,

JuvDisease ⊑ Disease ⊓ ∀affects.(Child ⊔ Teen),

JointDisease ⊑ Disease ⊓ ∃damages.Joint

Child ⊔ Teen ⊑ ¬Adult}

This TBox *contains modeling errors*:

Ontology Design

TBox $\boldsymbol{T}$ = {$\text{JuvArthritis} \sqsubseteq \text{Arthritis} \sqcap \text{JuvDisease}$,

$\text{Arthritis} \sqsubseteq \exists \text{damages}.\text{Joint} \sqcap \forall \text{damages}.\text{Joint} \sqcap \exists \text{affects}.\text{Adult}$,

$\text{JuvDisease} \sqsubseteq \text{Disease} \sqcap \forall \text{affects}.(\text{Child} \sqcup \text{Teen})$,

$\text{JointDisease} \sqsubseteq \text{Disease} \sqcap \exists \text{damages}.\text{Joint}$,

$\text{Child} \sqcup \text{Teen} \sqsubseteq \neg \text{Adult}$}

This TBox *contains modeling errors*:

- Juvenile arthritis is a kind of juvenile disease
  - Juvenile disease affects only children or teens, which are not adults
    - *A juvenile arthritis cannot affect any adult*
- Juvenile arthritis is a kind of arthitis
  - Each arthritis affects some adult
    - *Each juvenile arthritis affects some adult*

Concept Satisfiability

- What is the *impact of the error*?
  - All (interpretation) models ζ of TBox ***T*** must be such that φ(JuvArthritis) = ∅
  - A juvenile arthritis cannot exist!
  - We cannot add data concerning juvenile arthritis
- Such errors can be detected by solving the following problem:
  - Concept **satisfiability** w.r.t. a TBox:
  - An instance is a pair ⟨C, ***T***⟩ with C a concept and ***T*** a TBox. The answer is **true** iff a model ζ |= ***T*** exists such that φ(C) ≠ ∅.

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Concept Subsumption

TBox $\boldsymbol{T}$ = {JuvArthritis $\sqsubseteq$ Arthritis $\sqcap$ JuvDisease,

Arthritis $\sqsubseteq$ $\exists$damages.Joint $\sqcap$ $\forall$damages.Joint $\sqcap$ $\exists$affects.Adult,

JuvDisease $\sqsubseteq$ Disease $\sqcap$ $\forall$affects.(Child $\sqcup$ Teen),

JointDisease $\sqsubseteq$ Disease $\sqcap$ $\exists$damages.Joint,

Child $\sqcup$ Teen $\sqsubseteq$ $\neg$Adult}

Parts of our arthritis TBox $\boldsymbol{T}$, however, do *conform to our intuitions*:

- Juvenile arthritis is a kind of juvenile disease
  - Juvenile disease is a kind of disease
    - *Juvenile arthritis is a kind of disease*

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Concept Subsumption

TBox $\boldsymbol{T}$ = {JuvArthritis ⊑ Arthritis ⊓ JuvDisease,

Arthritis ⊑ ∃damages.Joint ⊓ ∀damages.Joint ⊓ ∃affects.Adult,

JuvDisease ⊑ Disease ⊓ ∀affects.(Child ⊔ Teen),

JointDisease ⊑ Disease ⊓ ∃damages.Joint,

Child ⊔ Teen ⊑ ¬Adult}

Parts of our arthritis TBox $\boldsymbol{T}$, however, do *conform to our intuitions*:

- Juvenile arthritis is a kind of juvenile disease
  - Juvenile disease is a kind of disease
    - *Juvenile arthritis is a kind of disease*
- Juvenile arthritis is a kind of arthritis
  - Each arthritis damages some joint
    - *Each juvenile arthritis damages some joint*

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Concept Subsumption

TBox $\boldsymbol{T}$ = {JuvArthritis ⊑ Arthritis ⊓ JuvDisease,

Arthritis ⊑ ∃damages.Joint ⊓ ∀damages.Joint ⊓ ∃affects.Adult,

JuvDisease ⊑ Disease ⊓ ∀affects.(Child ⊔ Teen),

JointDisease ⊑ Disease ⊓ ∃damages.Joint,

Child ⊔ Teen ⊑ ¬Adult}

Parts of our arthritis TBox ***T***, however, do *conform to our intuitions*:

- Juvenile arthritis is a kind of juvenile disease

  Juvenile disease is a kind of disease

  Juvenile arthritis is a kind of disease

- Juvenile arthritis is a kind of arthritis

  Each arthritis damages some joint

  Each juvenile arthritis damages some joint

- Conclusion: *Juvenile arthritis is a joint disease*.

Concept Subsumption

TBox $\boldsymbol{T}$ = {JuvArthritis $\sqsubseteq$ Arthritis $\sqcap$ JuvDisease,

Arthritis $\sqsubseteq$ $\exists$damages.Joint $\sqcap$ $\forall$damages.Joint $\sqcap$ $\exists$affects.Adult,

JuvDisease $\sqsubseteq$ Disease $\sqcap$ $\forall$affects.(Child $\sqcup$ Teen),

JointDisease $\sqsubseteq$ Disease $\sqcap$ $\exists$damages.Joint,

Child $\sqcup$ Teen $\sqsubseteq$ $\neg$Adult}

We have discovered new interesting information

All models $\zeta = (\delta, \varphi)$ of $\boldsymbol{T}$ must be such that φ(JuvArthritis) $\subseteq$ φ(JointDisease)

Juvenile arthritis is a sub-type of joint disease

All instances of juvenile arthritis are also joint diseases

Such implicit information is detectable by solving the following problem:

- Concept **subsumption** w.r.t. a TBox:
- An instance is a triple $\langle$C, D, $\boldsymbol{T}\rangle$ with C, D concepts and $\boldsymbol{T}$ a TBox. The answer is **true** iff φ(C) $\subseteq$ φ(D) for each $\zeta \models \boldsymbol{T}$ (written as $\boldsymbol{T} \models$ C $\sqsubseteq$ D).

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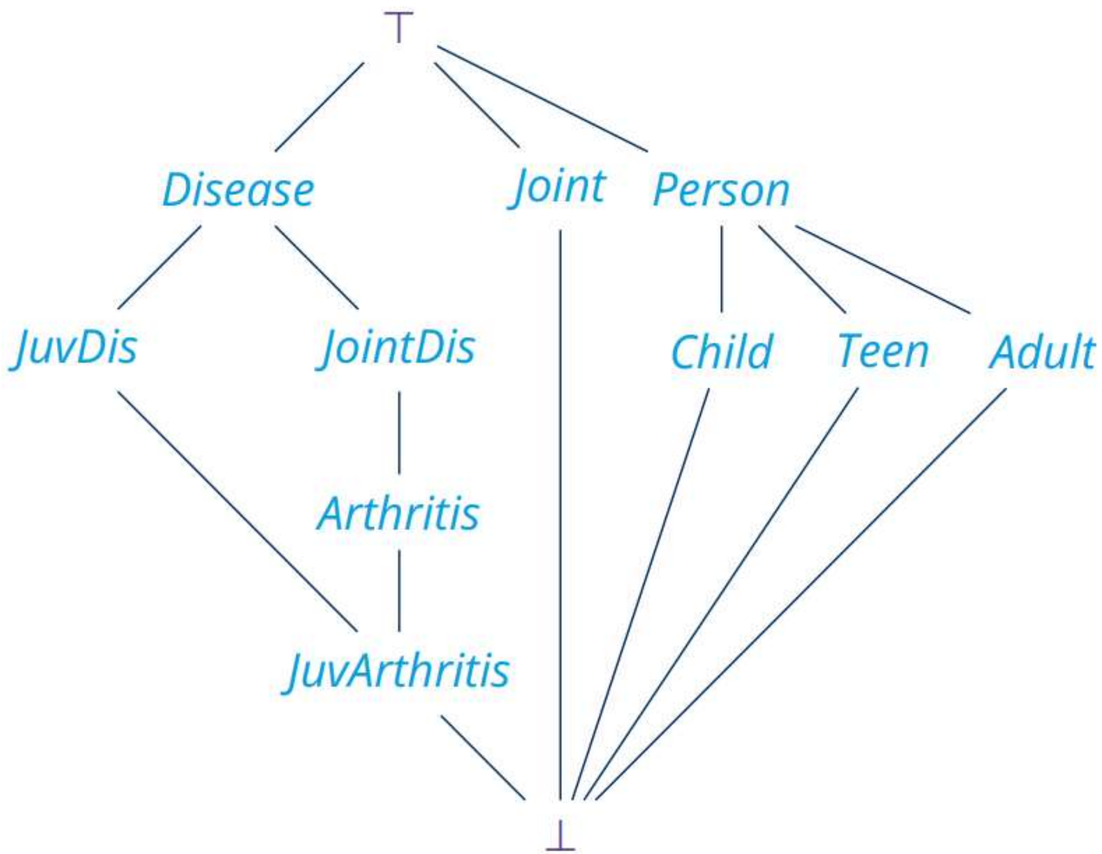
TBox Classification

Problem of finding all subsumptions between atomic concepts in ***T***.

Allows us to organise atomic concepts in a subsumption hierarchy:



Summary of Basic Reasoning Problems

Let Σ = (***T*** , ***A***) be an $\mathcal{ALC}$ knowledge base, C, D possibly complex $\mathcal{ALC}$ concepts, and b an individual / constant. We say that:

1. C is *satisfiable* with respect to ***T*** if there exists a model ζ of ***T*** and some d ∈ δ with d ∈ φ(C);
2. C is *subsumed by* D with respect to ***T***, written ***T*** |= C ⊑ D, if φ(C) ⊆ φ(D) for every model ζ of ***T***;
3. C and D *are equivalent* with respect to ***T***, written ***T*** |= C ≐ D, if φ(C) = φ(D) for every model ζ of ***T***;
4. Σ is *consistent* if there exists a model of Σ.
5. b is an instance of C with respect to Σ, written Σ |= C(b), if φ(b) ∈ φ(C) for every model ζ of Σ.

For clarity, we write $C \sqsubseteq_{\boldsymbol{T}} D$ for ***T*** |= C ⊑ D and $C \doteq_{\boldsymbol{T}} D$ for ***T*** |= C ≐ D.

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Reasoning Problem Reductions

All 5 previous basic problems can be reduced to finding **KB (in)consistency**.

Let $\Sigma = (\boldsymbol{T}, \boldsymbol{A})$ be an $\mathcal{ALC}$ knowledge base, C, D possibly complex $\mathcal{ALC}$ concepts, and b an individual / constant. We say that:

1. $C \doteq_{T} D$ iff $C \sqsubseteq_{T} D$ and $D \sqsubseteq_{T} C$;
2. $C \sqsubseteq_{T} D$ iff $C \sqcap \neg D$ is not satisfiable with respect to $\boldsymbol{T}$;
3. C is satisfiable with respect to $\boldsymbol{T}$ iff $C \not\sqsubseteq_{T} \perp$;
4. C is satisfiable with respect to $\boldsymbol{T}$ iff $(\boldsymbol{T}, \{C(b)\})$ is consistent.
5. $\Sigma \models C(b)$ iff $(\boldsymbol{T}, \boldsymbol{A} \cup \{\neg C(b)\})$ is not consistent.

All 5 basic problems **correspond** with its reasoning services, e.g.:

- (Satisfiability problem). Given a TBox $\boldsymbol{T}$ and a concept C, check whether C is satisfiable with respect to $\boldsymbol{T}$.

All can be realised via KB consistency checks, e.g.:

$$\Sigma \models C \sqsubseteq D \text{ iff } (\boldsymbol{T}, \boldsymbol{A} \cup \{C(a) \sqcap \neg D(a)\}) \text{ is not consistent}$$

for a an individual name not occurring in $\boldsymbol{A}$.

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Additional Reasoning Services

We can define additional reasoning services in terms of basic ones:

- *Classification* of a TBox: given a TBox $\boldsymbol{T}$, compute the subsumption hierarchy of all concept names occurring in $\boldsymbol{T}$. That is, for each pair C, D of concept names occurring in $\boldsymbol{T}$, check if

  $\boldsymbol{T} \models C \sqsubseteq D$ or $\boldsymbol{T} \models D \sqsubseteq C$;

- Checking the *satisfiability* of concepts in $\boldsymbol{T}$: given a TBox $\boldsymbol{T}$, for each concept name C in $\boldsymbol{T}$, test if

  $\boldsymbol{T} \not\models C \sqsubseteq \bot$;

- *Instance retrieval*: given a concept C and a knowledge base Σ, **return all those individual names** b such that b is an instance of C with respect to Σ. That is, for each individual name b occurring in Σ, check if

  $\boldsymbol{T} \models C(b)$;

- *Realisation* of an individual name: given an individual name b and a knowledge base Σ, **return all those concept names** C such that b is an instance of C with respect to Σ. That is, for each concept name C occurring in Σ, check if

  $\boldsymbol{T} \models C(b)$;

Summary

- Modern DL notations are there for its ease of use, readability, and standardisation across different DL variations.
- $\mathcal{AL}$ is often considered as the archetypical description logic, from which others are defined.
- $\mathcal{ALC}$ becomes the standard DL model, and the base of OWL and OWL 2.0.
- $\mathcal{ALC}$ Knowledge Bases provide various reasoning services where all can be reduced to finding KB (in)consistency.

Until the next session:

- Download Protégé (the one that doesn't need installing) to your computer.